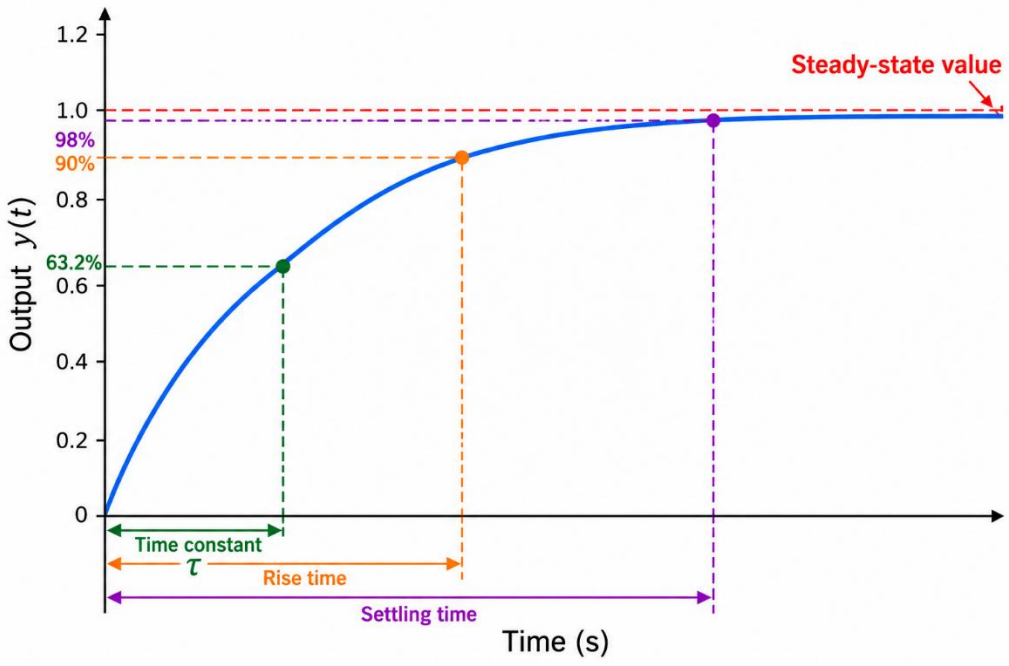
LAB NO: 5

Time Response of First Order Systems



Objectives

- Derive transfer functions from physical schematics and block diagrams.
- Derive first-order step responses analytically and verify them in MATLAB.
- Use tf, step, impulse, lsim, pole, dcgain, stepinfo and pzmap.
- Investigate gain K and time constant τ effects.
- Verify the 63.2% time-constant property and estimate settling time.
- Identify unknown parameters from a response curve.

Guided Tour: MATLAB Functions

Function	Purpose	Example
tf	Create transfer function	G = tf(K,[tau 1]);
step	Unit-step response	step(G);
impulse	Unit-impulse response	impulse(G);
lsim	Response to custom input	y = lsim(G,u,t);
pole	Find poles	pole(G)
dcgain	Steady-state gain	dcgain(G)
stepinfo	Time-response metrics	stepinfo(G)
pzmap	Pole-zero map	pzmap(G);
grid on	Improve plot readability	grid on;
hold on/off	Overlay plots	hold on; ... hold off;

Lab Experiment 1 — Basic Step Response

Analyze:

$$G(s) = 2/(5s + 1)$$

```
clc;
clear;
close all;
K = 2;
tau = 5;
G = tf(K,[tau 1]);
figure;
step(G);
grid on;
title('Unit-Step Response: G(s) = 2/(5s + 1)');
xlabel('Time (s)');
ylabel('Output y(t)');

system_pole = pole(G)
steady_state_gain = dcgain(G)
response_info = stepinfo(G)
```

Predict the final value, pole and time constant before running MATLAB.
Verify pole and DC gain using MATLAB.
Read the time at which the output reaches about 63.2% of its final value.
Use stepinfo to estimate rise time and settling time.
Explain why this ideal first-order response does not overshoot.

Lab Experiment 2 — Effect of Gain K

Keep $\tau = 4$ s and vary K = 0.5, 1, 2 and 4.

```
clc; clear; close all;
tau = 4;
K_values = [0.5 1 2 4];

figure; hold on;
for K = K_values
    G = tf(K,[tau 1]);
    step(G);
end
grid on;
legend('K=0.5','K=1','K=2','K=4','Location','southeast');
title('Effect of Gain K');
xlabel('Time (s)'); ylabel('Output');
hold off;
```

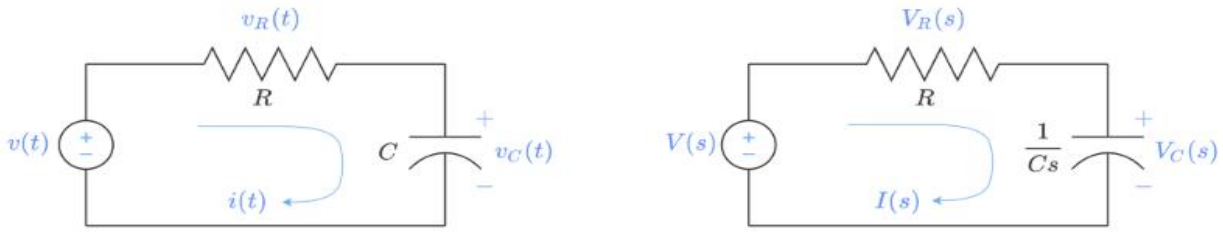
Record final value for each K.
Check whether pole and time constant change.
Explain why response magnitude changes while the time scale remains similar.

Control System- CP-402

Lab Exercise 1:

First order response characteristics of an RC circuit

The circuit below, shown in both the time domain and s-domain, is known as an RC circuit.



RC circuit shown in both the time domain and s-domain

V is the input voltage and V_C is the output voltage measured across the charging capacitor. The transfer function from $V(s)$ to $V_C(s)$ is $T(s) = \frac{V_C(s)}{V(s)}$.

The transfer function can be adjusted by changing the resistor, R, and capacitor, C, values. Nominal values for R and C are provided in the script below.

Write a MATLAB script that creates the system's transfer function and analyzes the system's transient response.

The MATLAB script must do the following:

1) Create the transfer function $T(s) = \frac{V_C(s)}{V(s)}$. The transfer function must be stored in the variable `T`

2) Calculate the system's time constant, T_C. The value of T_C must be stored in the variable `Tc`

3) Calculate the rise time, T_r. The value of T_r must be stored in the variable `Tr`

4) Calculate the settling time, T_s. The value of T_s must be stored in the variable `Ts`

5) Determine the resistor value (keeping the capacitor the same) that produces a settling time of $T_s = 0.1$ seconds. The resistor's value must be stored in the variable `R2`

6) Create the transfer function $T_2(s) = \frac{V_C(s)}{V(s)}$ using the resistor value, `R2`, determined in Step 5. The transfer function must be stored in the variable `T2`

```
% clear; close all; % clear all variables and graphs
%                    include when using the desktop version of MATLAB.
%                    clear and close all are not needed in MATLAB Grader online.
format compact;  % removes extra lines in the output

% Nominal system values
R = 25e3;
C = 10e-6;


% 1) Create the transfer function. Store in variable T


% 2) Calculate the system's time constant. Store in variable Tc


% 3) Calculate the rise time. Store in variable Tr


% 4) Calculate the settling time. Store in variable Ts


% 5) Determine the resistor value (keeping the capacitor the same) that
%    produces a settling time of 0.1 seconds. Store in variable R2


% 6) Create the transfer function using the resistor value, R2, determined in Step 5.
%    Store in variable T2


% Uncomment the following line to view the step responses to a 10V input.
% step(10*T); hold on; step(10*T2); title("Response to 10V step input"); legend("T(s)","T2(s)");

```

Lab Exercise 2:

RC Low-Pass Circuit

For a series resistor R and capacitor C, with output voltage across the capacitor: as give in Exercise 1,

$$RC\, dvout(t)/dt + vout(t) = vin(t)$$

$$(RCs + 1)Vout(s) = Vin(s)$$

Control System- CP-402

$$G(s) = Vout(s)/Vin(s) = 1/(RCs + 1)$$

1. A series RC circuit has R = 10 kΩ and C = 100 µF. Input is Vin and output is capacitor voltage Vout. Derive Vout(s)/Vin(s), identify K and τ, and predict the unit-step final value.
2. For a series RC low-pass circuit, R = 4.7 kΩ and C = 220 µF. Derive the transfer function and calculate the time to reach 63.2% of the final value.
3. Design a first-order system with unit-step final value 2.5 and approximate 2% settling time 8 s. Use Ts ≈ 4τ, then verify in MATLAB.